

Vision-Based Human Activity Recognition Utilizing DCNN and Stacked Multi-stage Multimodal Fusion Strategy

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Abstract—Human Activity Recognition (HAR) possesses significant potential in pattern recognition, machine learning, wearable computing, computer vision, and notably in rehabilitation assistance. With the ability to automatically track patients' recovery progress, it addresses the issue of insufficient human resources. Presently, improving recognition accuracy hinges on the fusion of multiple information sources like physiological signals, motion sensors, and visual sensors. In this study, a multi-stage multimodal fusion strategy based on stacked integration with Deep Convolutional Neural Network (DCNN) is proposed. Firstly, the inertial sensor and depth sensor data are converted into image data, and the data enhancement method is used to extract edge information using Prewitt, and features are extracted and spliced by the DCNN to realize multimodality at the feature layer. At the decision layer, the two unimodal decisions are fused using the stacking fusion strategy to obtain the final classification decision. The experimental results show that this strategy exhibits significant advantages in human motion recognition, with accuracy rates of 99.85% and 99.51% on Kinect V2 and UTD-MHAD datasets, respectively, exceeding the state-of-the-art deep learning methods on the same dataset.

Keywords—Sensors Fusion; DCNN; Rehabilitation Robots; Human Activity Recognition

I. INTRODUCTION

As the diseases' onset age lowers, hemiparesis becomes increasingly common in cardiovascular, cerebrovascular, and neurological conditions[1]. Despite the growing importance of regular rehabilitation, medications and surgery do not fully restore function in hemiplegic patients, and prolonged immobilization can lead to complications. With rehab resources lacking in China, much demand remains unmet, highlighting the relevance of a comprehensive rehabilitation robot system. This system would interpret the user's movement intentions and physical conditions, relay real-time feedback to the controller, and deliver essential assistance. Human motion intention recognition technology within this system optimizes user-robot synergy and comfort. The accuracy of this recognition significantly affects the robot's functionality and user experience, hence studying it bears considerable significance.

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Current human motion recognition techniques are mainly divided into visual and wearable sensor methods. Visual sensors, located within specific environments, include a variety of cameras like RGB-D, infrared, time-of-flight, multispectral, sprite, and thermal imaging. They recognize activities through pictures or surveillance videos with the help of image and video analysis technologies. The advent of cost-effective color depth detection devices, for example, Microsoft Kinect and Asus Xtion-Pro Live RGB-D cameras[2], has eased the acquisition of 3D and depth data. Depth sensor's proficiency for joint tracking has intrigued numerous computer vision researchers. For instance, Patwary[3] used multimodal RGB-D video data and a semi-supervised learning algorithm, FSSL-PAR, for bedridden patient activity recognition. The FSSL-PAR algorithm not only effectively recognized patient activities, but also substantially improved the classifier's generalization capability, outperforming other supervised learning algorithms. Batool et al.[4] proposed a K-Ary entropy classifier algorithm based on Probabilistic Boolean Incremental Learning (PBIL). The experiments employed depth data and RGB images of three public datasets, and utilized 2.5D point cloud modeling and ridge extraction techniques to extract full-body features. The results revealed a high accuracy of 95.05%, 95.56%, and 95.08%. While visual sensors recognize movements through image analysis, they face equipment or environment-related challenges. In contrast, wearable sensors, which work well in large spaces and start functioning when the wearer moves, are more flexible and thus commonly used in Human Activity Recognition (HAR). A new strategy was proposed [5] for constructing an information semantic space based on IMU(Inertial Measurement Unit) data for human activity recognition. The experiments incorporated a combination method of video-based and word-based semantic spaces to improve the zero-shot performance in human activity recognition using IMU data. The results suggested a significant potential in enhancing activity recognition tasks using IMU data, with the aid of videos of human activities and the related models developed in the computer vision space.

Systems solely using inertial sensors face limits such as electromechanical delay and random drift interference, which may reduce their precise activity recognition capabilities. Merging various types of sensors, like inertial ones prone to drift, with more resilient ones can offset these issues,

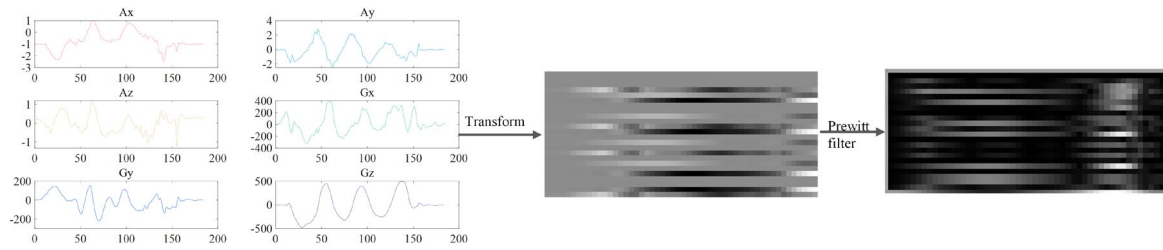


Fig. 1. The Preprocessing Process of IMU Signal Images

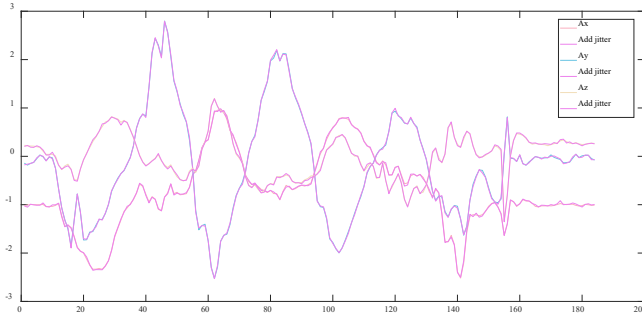


Fig. 2. Jitter Processing of Acceleration Sequences

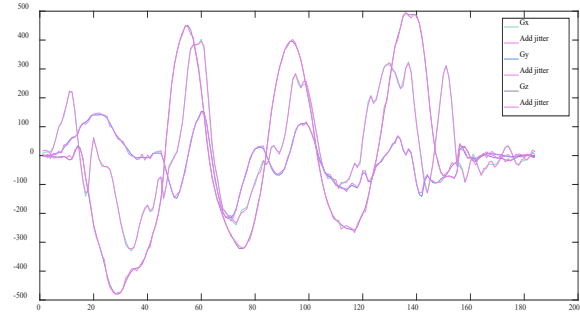


Fig. 3. Jitter Processing of Gyroscope Sequence

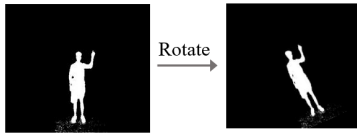


Fig. 4. The Processing of Depth Data Rotation

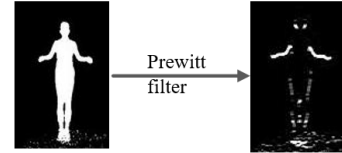


Fig. 5. Prewitt filtering process for depth data

enhancing accuracy. Recently, Human Activity Recognition (HAR) using mixed data has become popular, employing techniques such as probability, statistics, machine learning, and especially effective deep learning algorithms for automatic feature representation. In Bangaru[6] et al.'s study, EMG and inertial data were transmitted via Bluetooth from sensors on the arm to the network side, where they were fused and recognized using an ANN artificial neural network model. The model was able to recognize 15 scaffolding worker activities with 94% accuracy, thus providing an effective activity monitoring tool for the construction industry. In the Reference[7], Deep Belief Network (DBN) was applied to wearable sensor-based Chinese Sign Language (CSL) recognition. Experiments were conducted to collect electromyography(EMG), accelerometer, and gyroscope signals from eight participants as they completed 150 word gestures. The results show that feature fusion using network-generated features and DBN can achieve an optimal recognition accuracy of 95.1%, which provides a new approach for sensor fusion and CSL recognition.

In this study, a fusion strategy grounded on the multi-modality concept within the Stacking framework is introduced to explore the advantages of multi-modal fusion in human motion recognition. The primary achievements of the study are outlined below:

(1) A deep multimodal stacking fusion framework with a multilevel structure is employed for effective integration of diverse modalities. This ensures the full preservation and utilization of the uniqueness of each modality and enhances motion recognition accuracy.

(2) DCNN network model is utilized for data feature extraction and feature stitching experiments on different

layers, providing robust support for subsequent multimodal fusion.

(3) Experiments on public datasets demonstrate that the proposed stacking multimodal fusion surpasses other algorithms in recognition accuracy, underlining its superior performance.

The remainder of the paper includes a detailed explanation of the proposed method, experimental results, and a conclusion.

II. METHODS

The multimodal motion recognition procedure encompasses three parts: preprocessing of data, extraction of features, and fusion of sensors. Preprocessing involves data enhancement, conversion of sequence data into signal images, and edge feature extraction via Prewitt filters. Deep learning is exploited in the second stage to process sensor data, and concatenate extracted features. The ultimate phase utilizes a layered fusion technique for classification, where two unimodal features are run through a basic classifier and then a meta-classifier to yield the final motion recognition outcomes.

A. Data Preprocessing

The dataset used throughout the study is derived from the Kinect V2 Dataset public dataset [8]. In previous studies, the characteristics of human movement activities are usually extracted from each time-series sensor signal individually by manual means, which often ignores possible correlations between different signals, and the data volume is more dimensional, which makes the processing workload larger. In this study, we first convert the complex multidimensional raw sequence data into more easily processed image data. For the

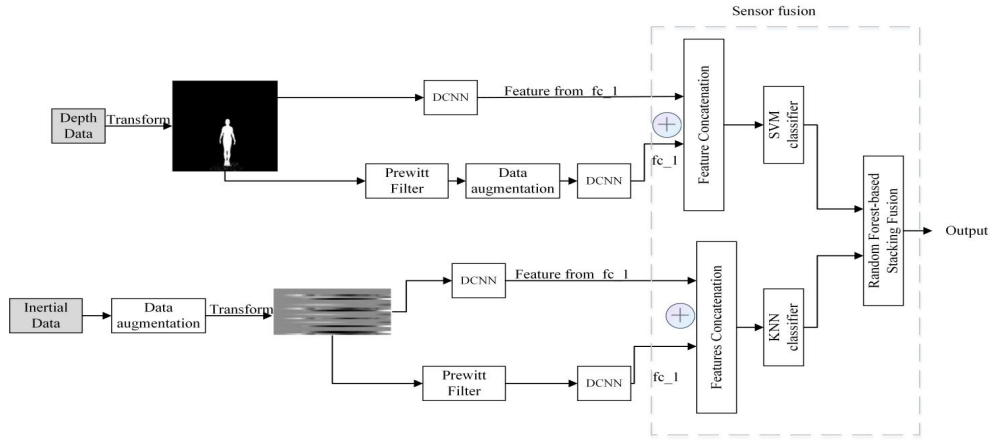


Fig. 6. Multimodal Stacking Fusion Framework

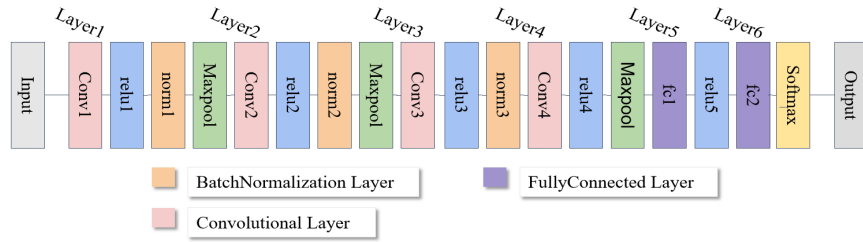


Fig. 7. DCNN Network Structure Diagram

raw inertial sensor data the method proposed in the literature [9] is used to stack the raw signals row by row in order to form a signal image(Fig.1), and the six sequences of rows are stacked in the following order: 12345613524625141536264, and the length of the sequences is 24. In the signal image, each signal sequence is given a chance to be adjacent to the other sequences adjacent to each other in order to train the network to be able to obtain the correlation of channels adjacent to each other. The frequency of the acquired data is 50 HZ, and the length of the signal image is determined as 52 to get the final image size of 24x52. For the depth data, since the original data is a three-dimensional sequence data, the third dimension data represents the number of frames of the converted image, and the first two dimensions represent the height-width pixel values of the image, which can be converted into a two-dimensional image by layer-by-layer extraction.

For the obtained image data, we apply Prewitt filtering to obtain image data that can reflect edge features(Fig.5). This is beneficial as these features can be complementary and distinctive during the subsequent feature extraction phase.

Considering the limited dataset and interference in signal acquisition, we augment the depth sensor and IMU sensor data to optimize the DCNN model training. For the depth sensor image data(Fig.4), we use rotation for data enhancement to accommodate environmental instabilities such as sensor inversion. For inertial sensor data(Fig.2 and Fig.3), we use a dithering strategy to introduce tiny random noise, including adding a certain amount of noise to the sensor readings and Gaussian noise, thus improving the model's resistance to interference.

B. Single-Modality Feature Extraction

1) Establishment and Training of DCNN Model

DCNN is a traditional CNN(Convolutional Neural Network) that adds depth to the network. This means that

DCNNs are able to extract and learn more complex and deeper features. Deeper neural networks typically have better performance for solving complex tasks, but also require greater computational resources and more sophisticated training strategies.

The architecture of the DCNN utilized in this study, illustrated in Fig.7, fundamentally consists of four convolutional layers, four pooling layers, and two dense layers. The number of kernel filters in each convolutional layer is sequential: 32, 64, 64, and 128, and the size of each filter is 3x3. After every convolutional layer, we have set a Rectified Linear Unit(ReLU) activation function layer and a Batch Normalization layer, followed by a pooling layer with a 2x2 kernel and a stride of 2. Batch Normalization's inclusion expedites model training, standardizes input layer data, improves layer-specific parameter learning, and accelerates network learning. After the pooling layers, we added two fully connected layers: one with 256 neurons and the other with the same number as the activity types in the dataset. The first layer maps high-level features into 256-dimensional space, while the second maps these to the output space. A softmax activation layer then generates category scores and classifies results. ReLU is used in training for input and hidden layers, addressing the vanishing gradient issue better than sigmoid and tanh. These enhancements boost the DCNN model's classification accuracy.

2) Feature concatenation

In the context of neural network applications, feature fusion typically involves extracting feature vectors from various layers or regions within the network. These vectors are then assembled to compile a more comprehensive feature vector. In the section related to feature concatenation, our main task focuses on the processing of feature data extracted via the DCNN. Firstly, attention is given to the Inertial Measurement Unit (IMU) data, which is either processed with

Prewitt filtering or left unprocessed. Secondly, depth data, either subjected to Prewitt filtering or left unprocessed, is also dealt with.

In DCNN, determining which layers of features should be fused is a critical step. Since the fusion of higher-level features helps to obtain more representative and descriptive features, we consider selecting from two fully connected layers for feature extraction and splicing. In addition, the Pearson correlation coefficient is used to evaluate and select the selected network layers with the following formula:

$$\text{corr}(X, Y) = \text{cov}(X, Y) / (\text{std}(X) * \text{std}(Y)) \quad (1)$$

X and Y are fully connected layer features of a single modality, $\text{std}(X)$ and $\text{std}(Y)$ denote the standard deviation of X and Y, and $\text{cov}(X, Y)$ is their covariance. Correlation coefficients with absolute values greater than 0.7, less than 0.3, or in between represent strong, weak, or moderate associations of the features, respectively.

C. Multi-modal Fusion for Action Recognition

This study proposes a deep multi-level feature fusion framework (Figure 6). The framework uses independent convolutional neural networks to process different modalities of data and employs multi-branch single-modal data concatenation to obtain more complementary features, thereby improving classification accuracy. Additionally, it utilizes a stack-based fusion strategy for multimodal fusion. This strategy includes two types of models: base models and meta-models. In this experiment, Random Forest, Support Vector Machine (SVM), and k-nearest Neighbors (KNN) are selected as meta-models because they have been proven to produce excellent overall performance in other experiments[10], and their performance in HAR tasks surpasses other classifiers. Specifically, we perform separate dual-branch processing for the depth dataset and the inertial sensor dataset, each using a neural network trained with a DCNN model. In one branch, we first utilize Prewitt filters for edge detection processing, then incorporate data enhancement

techniques such as rotation and perturbation to augment the data and improve the robustness and generalization of the model, and finally feed the processed data into the network.

Finally, the features obtained from the training of the two branches are spliced. Similarly, the inertial sensor dataset is processed in the same way. Subsequently, we perform classification prediction on the spliced deep features and inertial sensor features by the base model, respectively. Eventually, the predictions are fed into a meta-classifier to produce the final prediction.

III. EXPERIMENTS

A. Experimental Setup

The Kinect V2 action dataset used in the experiments is a publicly available dataset containing depth and inertial sensor data. The dataset consists of 10 actions performed by 6 subjects, with each subject repeating the action 5 times. The labels of these 10 actions can be known from the horizontal and vertical coordinates in the Fig12. In the experiment, we randomly divide the dataset into training data and test data in a 9:1 ratio. This ensures there is sufficient data for the model to learn various patterns and features, while also providing enough data for a preliminary estimate of model performance. Since the IMU sensor data is relatively homogeneous with insufficient features and a small amount of data, the IMU sequences are converted to images and processed using dithered data enhancement methods to prevent overfitting. Similarly, in order to increase the generalization ability of the trained model, the depth sensor data is processed using rotated data enhancement techniques. In addition, to generate additional features, we use Prewitt edge detection filtering for both IMU and depth data, which enriches the spliced features and contributes to the subsequent multimodal fusion classification accuracy. The platform used for the experiments is MATLAB 2020b, and the hardware resource is GPU.

B. Experiment Evaluation

1) Single-modal Feature Recognition Experiment

We utilize the depth data and IMU data preprocessed from the previous section, which includes both data sets that have and have not undergone Prewitt filtering. By conducting

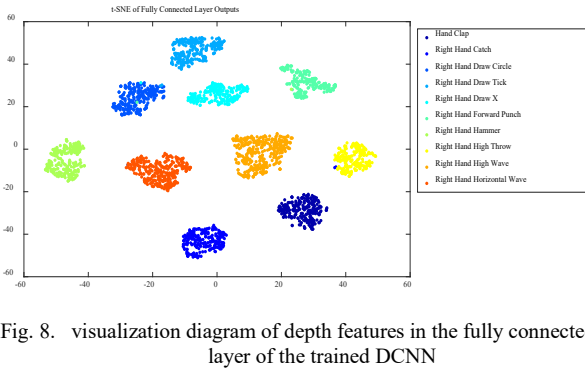


Fig. 8. visualization diagram of depth features in the fully connected layer of the trained DCNN

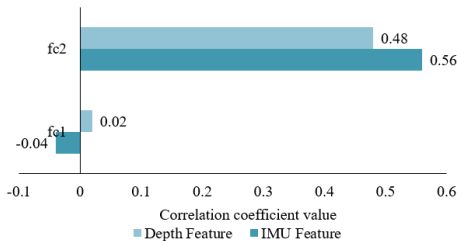


Fig. 9. comparison of feature correlation between fc1 and fc2 Layers

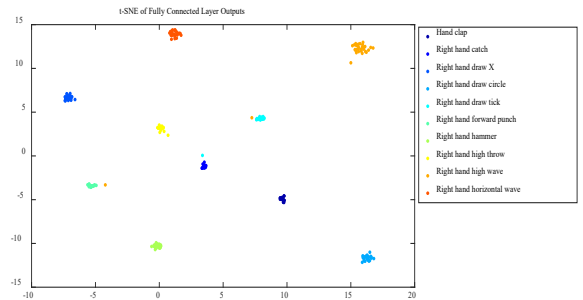


Fig. 10. visualization diagram of IMU features in the fully connected layer of the trained DCNN

TABLE I. EVALUATION INDICATORS FOR THE COMBINATION OF THREE CLASSIFIERS

	Accuracy	Precision	recall	F1-score
Combination1	99.85%	99.85%	99.85%	99.85%
Combination2	96.42%	96.59%	96.60%	96.59%
Combination3	96.14%	96.40%	96.38%	96.39%

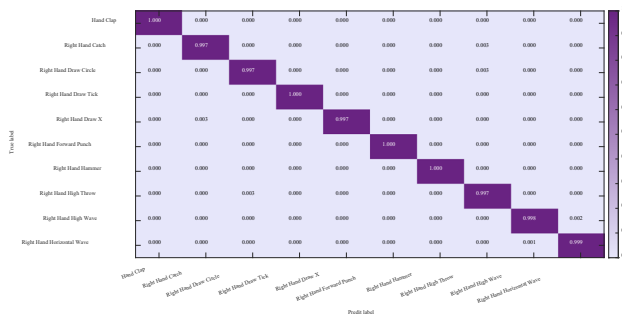


Fig. 12. confusion matrix of human motion recognition obtained through stacking fusion strategy on Kinect V2 dataset

training on the single sensor data using a neural network model, we are able to carry out feature extraction.

In this experiment, we explore the optimal training parameters for the DCNN network. The corresponding parameters are as follows: we employ Stochastic Gradient Descent with Momentum (SGDM) as the optimizer, and set the momentum parameter to 0.9. L2 regularization is chosen as the regularization method. For depth sensor data, the weight is set to 0.004, and for IMU data, it is set to 0.02 to prevent overfitting. We set the initial learning rate for depth sensor data training at 0.005, and for IMU data, it is set at 0.001. For IMU data training, the minimum batch size is set at 32, and for depth sensor data, it is set at 64. Training will be stopped early if the validation loss plateaus (patience=5).

In order to demonstrate more intuitively the performance of DCNN networks on feature classification, we visualize the fully connected layer neuron features using t-distributed Stochastic Neighbor Embedding (t-SNE) [11]. Fig. 8 and Fig. 10 illustrate the distribution of features of different behavioral categories, identified by different colors, in the feature space.

TABLE II. COMPARISON OF RECOGNITION ACCURACY OF DIFFERENT MODALITY COMBINATION MODELS

Previous methods	Data	Accuracy(%)
Chen et al. [12]	Kinect V2	98.3
Bulbul et al. [13]	UTD-MHAD	88.4
N.Dawar et al. [14]	UTD-MHAD	92.8
Mahjoub et al. [15]	UTD-MHAD	98.4
Performance of our methods		
IMU	Kinect V2	73.83
IMU+ Augmentation	Kinect V2	94.02
IMU+ Augmentation+Prewitt+ Concatenation	Kinect V2	94.80
Depth	Kinect V2	96.24
Depth+ Augmentation	Kinect V2	97.26
Depth+ Augmentation+Prewitt+ Concatenation	Kinect V2	98.38
Stacking	Kinect V2	99.85(17ms)
Stacking	UTD-MHAD	99.51(27ms)

Features of the same class of behavior are clustered together, while those of different categories are relatively separate. The occasional points far away from the same color and close to different colors may be instances of model misclassification. For data of two modalities, the features of different behaviors extracted by the DCNN network can be better distinguished, with very few overlaps of points of

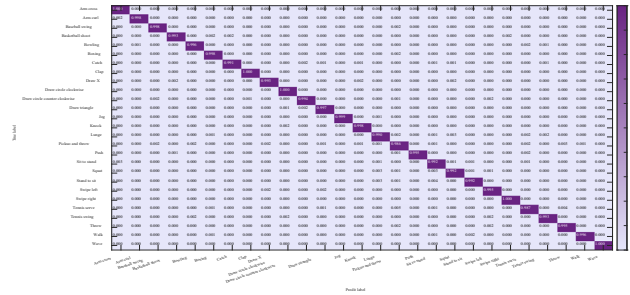


Fig. 11. confusion matrix of human motion recognition obtained through stacking fusion strategy on UTD-MHAD dataset

different colors. This suggests that the DCNN network is effective in feature extraction.

In MATLAB, the function corrcoef, which computes the Pearson correlation coefficient, is used to calculate the correlation between the Prewitt filter signal features used for the concatenation of layers fc1 and fc2 and the signal features that have not been filtered. The experimental results are shown in Fig. 9. As can be inferred from the data in the figures and tables, whether for Depth data or IMU data, the correlation of the concatenated signal features in the fc1 layer is smaller, thus layer fc1 is more suitable for extracting concatenated features.

2) Comparative Experiment on Different Classifier Combinations

In the selection of classifiers, this experiment uses SVM, KNN and Random Forest as either meta-classifiers or base classifiers. The goal is to compare the recognition accuracy under different classifier combinations and thus select the optimal scheme. The experiment initially applies KNN, SVM, and Random Forest classifiers for single-modal feature classification. As per the experimental results, the KNN classifier exhibits the best recognition effect for IMU features, reaching up to 94.8%. The SVM classifier excels in recognizing Depth features, achieving 98.31%, but its performance on IMU features is subpar. Conversely, the Random Forest classifier performs well on both Depth and IMU features, with classification accuracy rates hitting 88.9% and 97.87%, respectively.

Considering the classification effect of the above unimodal features, the following three main combination schemes are considered for the classifier of multimodal fusion for Stacking:

- Combination 1: KNN is used for IMU features, SVM is used for Depth features, and Random Forest is used as the meta-classifier.
- Combination 2: KNN is used for both IMU and Depth features, and Random Forest is used as the meta-classifier.
- Combination 3: Random Forest is used for IMU features, SVM is used for Depth features, and KNN is used as the meta-classifier.

As shown in the figure, the confusion matrix of Combination 1 (Fig. 11) for the recognition of ten kinds of movements has the largest values on the diagonal, representing the maximum number of positive instances and the highest accuracy of motion recognition, reaching 99.71%. Furthermore, in the comparative charts (Table I) for accuracy,

recall, precision, and F1-score, it can be seen that all evaluation indicators of Combination 1 are slightly higher than the other two combinations. Therefore, Combination 1 has the optimal activity recognition effect. Therefore, this study adopts a multi-modal fusion strategy with the first combination.

3) Comparison Experiment of Multi-modal and Single-modal Recognition

The main objective of this section is to explore the differences in feature recognition between multi-modal and single-modal approaches. The experiment compares the recognition accuracy on IMU and depth sensor data and tests the effects of data augmentation and Prewitt features concatenation in improving recognition accuracy. At the same time, it observes and compares whether data fusion could further improve the accuracy of recognition. In order to further verify the robustness of the algorithm, other datasets are used for validation, and comparisons are made with past studies that used the same datasets for human motion recognition.

From the experimental results (Table II), it can be seen that for the Kinect V2 dataset, the recognition accuracy of our proposed unimodal training model is 73.83% on a single IMU data. When data enhancement is performed on the IMU data, the model recognition accuracy is improved to 94.02%. After using data enhancement, splicing preprocessing technique, and the KNN classifier, the recognition accuracy of the IMU data is further improved to 94.8%. Our proposed unimodal training model on single-depth sensor data has a recognition accuracy of 96.24%. When the depth data is subjected to data augmentation, the recognition accuracy improves to 97.26%. After using data augmentation, splicing preprocessing technique, and SVM classifier, the recognition accuracy of the depth data further improves to 98.38%. Finally, after performing our proposed Stacking multi-sensor fusion model on the entire dataset, the recognition accuracy is further improved to 99.85%.

To verify the effectiveness of the model, we applied our proposed Stacking multi-modal fusion framework to the UTD-MHAD[16] dataset. The final classification recognition accuracy was 99.51%, and the resulting classification confusion matrix is shown in Fig.12. The table presents the fusion framework's inference rates for two datasets in microseconds (ms), indicating the time required for the classifier to process each test sample.

This experiment demonstrates the considerable advantage of the proposed multimodal recognition framework in accuracy, in comparison to unimodal recognition. The addition of data augmentation and feature concatenation techniques further strengthens the framework's recognition accuracy. Notably, this framework surpasses the accuracy of its predecessor, marking a substantial enhancement.

IV. CONCLUSION

In this paper, we innovatively propose a human motion recognition method using a deep multimodal stacked integrated fusion framework, where the raw sequence data of KinectV2 is converted into images and the model is optimized by data augmentation and the addition of Prewitt filtering features. A DCNN network model is used to extract and fuse features from preprocessed unimodal data. Experimental

results clearly show that our model exhibits high efficiency in extracting discriminative features. Our chosen stacking fusion strategy uses SVM, KNN, and Random Forest as fusion classifiers, and different combinations of classifiers were evaluated to determine the optimal model. Subsequently, we perform performance tests on the KinectV2 and UTD-MHAD datasets. The results show that for the same dataset, the recognition accuracy of our proposed Stacking multimodal fusion exceeds that of other algorithms, highlighting the superior performance of our approach.

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